Condition-Based Maintenance Optimization for Multicomponent Systems Under Imperfect Repair—Based on RFAD Model

Hong-Guang Ma, Ji-Peng Wu, Xiao-Yang Li, and Rui Kang

Abstract—**Condition-based maintenance has been developed as a very efficient strategy for guaranteeing multicomponent system performance and preventing unexpected failures. However, there are shortcomings in the existing condition-based maintenance optimization models. First, the existing models do not utilize the accelerated degradation testing (accelerated degradation testing) data obtained at the stage of component development. Second, most of these models assume perfect repair instead of imperfect repair. Third, the degradation models used in these condition-based maintenance models cannot consider the epistemic uncertainty. Motivated by these problems, this paper presents a new condition-based maintenance optimization model for multicomponent systems with imperfect repair. An integrated degradation prediction framework utilizing both ADT data and field data is presented to timely update the parameters in the proposed model. In order to solve the proposed multivariable, nonlinear programming model, a novel genetic algorithm with self-crossover operation and shift-mutation operation is developed. Numerical examples and comparisons are conducted to evaluate the performance of the proposed model. Results show that the proposed model can evaluate the degradation process of components accurately and achieve lower total maintenance cost.**

Index Terms—**Condition-based maintenance, genetic algorithm, imperfect repair, multicomponent systems.**

I. Introduction

MAINTENANCE is necessary for guaranteeing system performance and preventing unexpected failures [1], [2]. One of the most popular preventive maintenance policies is scheduled maintenance, whereby pre-establishes the routines and time interval of the maintenance. In recent years, condition-based maintenance has been developed as a very efficient maintenance strategy [3]. Within the condition-based maintenance framework, the health condition of a piece of equipment is monitored by collecting and analyzing data, such as vibration data, acoustic emission data, oil analysis data, and temperature data, and optimal maintenance actions are scheduled for preventing equipment breakdown and reducing total operation costs.

Condition-based maintenance focuses on scheduling maintenance activities by periodically or continuously monitoring specific measures that are related to the health and performance of the systems that are being maintained. Once these measures exceed a predefined failure threshold, the system is shut down for maintenance. There are plenty of literatures on the condition-based maintenance models for single-unit systems, which can be classified into two categories: First, condition-based maintenance models for the discrete-state degradation processes; and second, condition-based maintenance models for the continuous-state degradation processes [4]. For example, Nakagawa and Ito [5] presented condition-based maintenance policies for critical components of fossil fired power plants. Failure occurs when the cumulative damage of the component exceeds a predetermined damage level. Sloan and Shanthikumar [6] assumed that the condition of equipment is monitored at equidistant time intervals. The authors also assumed that the equipment can fail within an inspection interval, with exponential distribution of failure times. Wu *et al.* [7] developed an integrated neural-network-based decision support system for predictive maintenance of rotational equipment. The integrated system focused on minimizing the expected cost per unit operational time, using a heuristic managerial decision rule for different scenarios of predictive and corrective cost compositions.

The above reported studies mainly focus on a single unit or component. The preventive and replacement maintenance decisions are made individually on the components, based on their ages, condition monitoring data, and a given condition-based maintenance policy. However, with the development of multifunction and complex structure of modern equipment, the maintenance strategy treating the equipment as a whole system has been difficult to meet the diversified maintenance requirements of complex equipment. In determining the maintenance

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strategy of a multicomponent system, the situation of each component needs to be taken into consideration. For example, the fixed preventive or replacement cost is incurred for sending a maintenance team to the affected site, for a preventive or replacement maintenance action of one component. Then, it would be more economical to perform preventive or replacement maintenance actions on multiple components at the same time.

Nowadays, more and more researchers have been focusing on models of the maintenance optimization for multicomponent systems. Marseguerra *et al.* [8] considered continuously monitored multicomponent system and used a genetic algorithm for determining the optimal degradation level. Archibald and Dekker [9] developed a modified block replacement policy for multicomponent systems, where preventive replacement is performed at constant intervals and only on the components that are older than a prespecified age limit. Schouten and Vanneste [10] proposed two simple control policies for multicomponent systems by considering several intermediate states. Castanier *et al.* [11] presented a condition-based maintenance policy with nonperiodic inspections for a two-unit series system, and developed a stochastic model based on the semiregenerative properties of the maintained system state for cost evaluation. Chen *et al.* [12] investigated the optimal condition-based maintenance policy when the system degradation conforms to the inverse Gaussian process with random-drift model. The random drifts were used to represent heterogeneities commonly observed across the product population. However, they assumed that both corrective and preventive maintenance are perfect, restoring the unit to a as-good-as-new state. Some other studies on multicomponent system maintenance are summarized by Keizer *et al.* [13] and Wang [14].

The previous works model the degradation processes of components or systems by a certain theoretical degradation model with fixed parameters. However, in practical applications, the degradation processes of components may change in time. Therefore, the parameters in the degradation model should be correspondingly adjusted according to the actual degradation of components. Furthermore, perfect preventive maintenance is usually handled in pervious studies of maintenance policies. That is, after a preventive maintenance action, the component is completely restored and considered as good as new. However, this is unrealistic in practice, and the system condition after a preventive maintenance action is somewhere between good as new and bad as old, which is called imperfect maintenance. The research on imperfect maintenance for condition-based maintenance of multicomponent systems has not received enough attention.

In this paper, we propose a condition-based maintenance optimization model for multicomponent systems with consideration of imperfect maintenance. The objective is to minimize the total cost of an imperfect condition-based maintenance strategy, which includes the preventive or replacement maintenance costs and the loss cost incurred by the downtime of the system. The degradation processes of the components are described by the random fuzzy accelerated degradation (RFAD) model proposed by Li *et al.* [15], which can simultaneously consider the time-stress-dependent structure, the random uncertainties caused by random effects in time dimension and unit-to-unit variations, and the epistemic uncertainty caused by the small sample problem. In order to keep up with the timely change in the degradation processes, the parameters in the RFAD model are continuously updated based on the accelerated degradation testing (ADT) data and the real-time monitored data.

This paper include the following contributions.

1) This paper proposes a condition-based maintenance optimization model for multicomponent systems based on the RFAD model.
2) An integrated degradation prediction framework using both ADT data and real-time monitored data is presented to periodically update the parameters in the proposed model, which is unique in literature.
3) Since the proposed model is computationally expensive, a novel genetic algorithm with self-crossover operation and shift-mutation operation is developed.
4) Numerical examples and comparisons are conducted to highlight the advantages of the proposed model in depicting the degradation of components more accurately, more fully utilizing the capacity of the components, and achieving lower total maintenance cost.

The remainder of the paper is organized as follows. Section II is devoted to the description of the multicomponent system under consideration and the formulation of the maintenance optimization model. A genetic algorithm with self-crossover operation and shift-mutation operation is developed in Section III. Section IV conducts numerical examples and comparisons to evaluate the performance of the proposed model. Section V provides conclusions and some possible future research directions.

II. Problem Description and Formulation

We consider a series system consisting of M repairable components under inspection. The down or failure of any component will lead to the down or failure of the whole system. The operation horizon contains a total of N inspections. The inspection methods may include vibration monitoring, oil analysis, etc., depending on the specific problem. The inspections can obtain the current degradation level of each component.

For each component $m \in M$, its degradation level will be strictly increasing without maintenance actions. It will be a great probability of failure and cannot work normally, once its degradation level exceeds a threshold $\omega_m, m \in M$. The threshold ω_m can be seen as a degradation level, which must not be exceeded for economical or security reasons. Therefore, after each inspection, a decision must be made for each component among three possible actions: preventive maintenance, replacement maintenance, or wait until next inspection. If a preventive or replacement maintenance action is performed, a maintenance cost will be incurred, which includes two parts: the preventive or replacement maintenance cost and the loss cost incurred by the downtime of the system. Here, opportunistic maintenance is considered for the multicomponent system, which means that when more than one components are performed on the maintenance actions simultaneously, the downtime of the system is the longest maintenance time among the components, rather than the sum of the maintenance times of all components. Therefore, the problem is how to plan the maintenance actions of all components to minimize the total maintenance cost with no failure of all components.

The following assumptions are made regarding the multicomponent system.

1) The M repairable components are all new at the initial time, that is, the degradation level of each component at the initial time is 0.
2) The stochastic dependence is not considered, i.e., the degradation process of each component is independent of other components [16], [17].
3) We focus on the maintenance optimization, so the inspection interval is not a design variable of the optimization problem. And the inspections are assumed to be performed at zero costs.

In the following, we first introduce the RFAD model to depict the degradation process of components, and present an integrated method that utilizes both the ADT data and the real-time monitored data to update the model parameters. Then, a condition-based maintenance optimization model is proposed for the multicomponent system.

A. Degradation Model and Parameters Updating

In the maintenance optimization, one important issue is to choose an appropriate model to depict the degradation process of the components. In this paper, the RFAD model proposed by Li *et al.* [15] is first introduced to the field of maintenance optimization to describe the degradation processes of components. The RFAD model considers the time-stress-dependent structure, the random uncertainties caused by random effects in time dimension and unit-to-unit variations, and the epistemic uncertainty caused by the small sample problem simultaneously and can be expressed as follows:

$$\widetilde{Y}(t) = \widetilde{d}\,(S)\,t + \sigma_B B\,(t)\,,$$
$$\ln \widetilde{\mu} = \widetilde{a} + \widetilde{b}\varphi\,(S)\,, \ln \sigma = u + v\varphi(S) \tag{1}$$

where $\widetilde{Y}(t)$ is the random fuzzy performance degradation process; $\widetilde{d}\,(S) \sim N(\widetilde{\mu},\, \sigma^2)$ is a random fuzzy variable; $\widetilde{\mu}$ is a fuzzy variable that represents the time-stress-dependent structure and the small sample problem; σ is a crisp value that represents the unit-to-unit variations; $\widetilde{a}$, $\widetilde{b}$, u, and v are acceleration model parameters; σ_B is the diffusion coefficient, which is constant; t is time and $B(t)$ is a standard wiener process, which is normally distributed with mean 0 and variance t, for each $t \in (0, +\infty)$; $\sigma_B B(t)$ represents the random effects caused by external disturbance in time dimension.

In practical applications, the actual degradation process of a component may change during each two inspections. Therefore, with more real-time monitored data collected, the parameters in the degradation model should be modified at each inspection, so as to describe the real degradation process more precisely. Motivated by this reason, we present an integrated method that utilizes both the ADT data and the monitored data to timely update the model parameters. To the best of our knowledge, this is the first time in the literature.

TABLE I
NOTATIONS

Notations	
N	the number of inspections, $n \in N$
$\overline{N}$	the maximum number of preventive maintenance actions for a same component
d_n	the collected monitored data at the n-th inspection
$\hat{n}$	the number of preventive maintenance actions before replacement
y_{ADT}	the ADT data
y_{normal}	all the monitored data collected currently
Ω_0	parameter estimations of the RFAD model (only use y_{ADT})
Ω_n	updated parameter estimations of the RFAD model (use y_{ADT} and y_{normal} after the n-th inspection)
Y_n	the degradation level at the n-th inspection
γ	the prespecified chance level

Suppose Y_n is the degradation level at the nth inspection, and the inspection interval is Δt. Based on the updated parameters, the cumulative distribution function of the degradation level at the $(n+1)$th inspection can be obtained as follows:

$$\tilde{F}_{Y_{n+1}}(y) = \Phi\left(\frac{Y_n + y - \exp\left(\tilde{a} + \tilde{b}\varphi\,(S_0)\right)\Delta t}{\exp\left(u + v\varphi\,(S_0)\right)\Delta t^2 + \sigma_B^2\Delta t}\right) \tag{2}$$

where S_0 is the stress level under normal operating conditions.

In order to ensure that the component will not be down between two inspections, it needs to judge whether a preventive or replacement maintenance should be performed at each inspection. Therefore, this paper adopts the following judgment criteria.

The chance that the degradation level at the next inspection will not exceed the failure threshold ω should be bigger than $\gamma, 0 \leq \gamma \leq 1$, that is

$$\mathrm{Ch}\{Y_{n+1} \leq \omega\} = \Phi\left(\frac{(\omega - Y_n) - \exp\left(\tilde{a} + \tilde{b}\varphi\,(S_0)\right)\Delta t}{\exp\left(u + v\varphi\,(S_0)\right)\Delta t^2 + \sigma_B^2\Delta t}\right) \geq \gamma \tag{3}$$

where γ represents the occurring possibility of $Y_{n+1} \leq \omega$ and is usually set to 0.9 or 0.95. In the numerical examples, we set $\gamma = 0.95$.

Using the notations defined in Table I, the procedure of maintenance optimization with parameter updating for a single component is listed in Algorithm 1.

Algorithm 1 An Integrated Method for Parameters Updating.

Step 1. Use the ADT data y_{ADT} and the RFAD model to obtain the initial estimations of the model parameters Ω_0, and move to **Step 2**.
Step 2. Input N, $\overline{N}$, γ, set $n = 1$, $\hat{n} = 0$, $y_{\mathrm{normal}} = []$ and move to **Step 3**.
Step 3. Collect d_n monitored field data, add them to y_{normal}, and move to **Step 4**.
Step 4. Compare n with N, if $n \leq N$, move to the **Step 5**; if not, the procedure is ended.
Step 5. Use the ADT data y_{ADT} and the field data y_{normal} to update the model parameters of the RFAD model (Ω_n), and move to **Step 6**.
Step 6. Calculate $\mathrm{Ch}\{Y_{n+1} \leq \omega\}$, and compare it with the prespecified chance level γ. If $\mathrm{Ch}\{Y_{n+1} \leq \omega\} \geq \gamma$, no preventive maintenance or replacement is needed, $\hat{n} = \hat{n}$, $n = n+1$ and move to **Step 3**; otherwise, move to **Step 7**.
Step 7. Compare $\hat{n}$ with $\overline{N}$, if $\hat{n} \leq \overline{N}$, a preventive maintenance action is needed, $\hat{n} = \hat{n}+1$, $n = n+1$, and move to **Step 3**; if not, a replacement action is needed, $\hat{n} = 0$, $n = n+1$, and move to **Step 3**.

B. Maintenance Optimization Model for Multicomponents Systems

Two binary variables, σ_{mn} and δ_{mn}, $m \in M$, $n \in N$, are defined to formulate the maintenance decision, as follows:

$$\sigma_{mn} = \begin{cases} 1, & \text{if a preventive maintenance is performed on} \\ & \text{component } m \text{ after the } n\text{th inspection} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\delta_{mn} = \begin{cases} 1, & \text{if a replacement maintenance is performed on} \\ & \text{component } m \text{ after the } n\text{th inspection,} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Denote Y_{mn} as the degradation level of component m before maintenance at the nth inspection and Y^*_{mn} as the degradation level of component m after maintenance at the nth inspection. If there is no maintenance action performed at the nth inspection, the degradation level of component m will not change, that is

$$Y^*_{mn} = Y_{mn}. \tag{6}$$

If a replacement maintenance action is performed at the nth inspection, the component becomes a new one, that is

$$Y^*_{mn} = 0. \tag{7}$$

If a preventive maintenance action is performed at the nth inspection, the component cannot be completely new, while the degradation level of component m is pulled back to some specified level between 0 and Y_{mn}. The degradation level of component m after the preventive maintenance can be calculated as

$$Y^*_{mn} = (1 - r_{m\hat{n}})Y_{mn} \tag{8}$$

where $r_{m\hat{n}} \in (0, 1)$ is the effect of the preventive maintenance action. Generally, the effect of the preventive maintenance action for a component decreases with the increase of the number of maintenance actions. $\hat{n}$ represents the number of preventive maintenance actions after the latest replacement of component m, which can be determined as follows:

$$n^* = \begin{cases} \max\{\underset{1 \leq n' < n}{\operatorname{argmax}}\, \delta_{mn'}\}, & \sum_{\theta=1}^{n} \delta_{m\theta} > 0, \\ 0, & \sum_{\theta=1}^{n} \delta_{m\theta} = 0, \end{cases}$$

$$\hat{n} = \sum_{\theta=n^*+1}^{n} \sigma_{m\theta}.$$

Therefore, the relationship between Y_{mn} and Y^*_{mn} can be formulated as follows:

$$Y^*_{mn} = Y_{mn} - r_{m\hat{n}} Y_{mn} \sigma_{mn} - Y_{mn} \delta_{mn}. \tag{9}$$

If a maintenance action is performed on component m at the nth inspection, a maintenance cost is incurred. The maintenance cost includes two parts: the preventive or replacement maintenance cost and the loss cost incurred by the downtime of the system. If component m is replaced at the nth inspection, the replacement cost is the purchase price of a new component m, denoted as p_m, plus a fixed start-up cost c_0. If a preventive maintenance action is performed on component m at the nth inspection, the maintenance cost C_{mn} and spending time T_{mn} are depending on both the effect of preventive maintenance and the system condition at the inspection. Without loss of generality, the relationship between the reduced degradation level and the maintenance cost can be given as a linear equation, as follows:

$$C_{mn} = (\alpha_m r_{m\hat{n}} Y_{mn} + c_0)\sigma_{mn} + (p_m + c_0)\delta_{mn} \tag{10}$$

where c_0 is the fixed startup cost for each maintenance action and α_m is a known proportional constant. Similarly, the relationship between the reduced degradation level and the time spent of the preventive maintenance, T_{mn}, can be given as a linear equation as follows:

$$T_{mn} = (\beta_m r_{m\hat{n}} Y_{mn} + t_0)\sigma_{mn} + (t_m + t_0)\delta_{mn} \tag{11}$$

where t_0 is the fixed startup time for each maintenance action, t_m is the time required for a replacement maintenance of component m, and β_m is a known proportional constant.

When more than one components are performed on the maintenance actions simultaneously at the nth inspection, the downtime of the whole system should be the longest maintenance time of the components, that is

$$T_n = \max_{1 \leq m \leq M} T_{mn}. \tag{12}$$

Assume the unavailability cost rate of the system is c_d. Thus, the loss cost incurred by the downtime of system at the nth inspection is

$$L_n = c_d T_n. \tag{13}$$

Therefore, the total maintenance cost of the system in the operation horizon can be calculated by

$$C = \sum_{m=1}^{M} \sum_{n=1}^{N} C_{mn} + \sum_{n=1}^{N} L_n. \tag{14}$$

Denote $Y_{n,n+1}^m$ as the degradation level of component m between the nth inspection and the $(n+1)$th inspection. Then, it should be

$$Y_{m,n+1} = Y_{mn}^* + Y_{n,n+1}^m, \;\; n = 1, 2, \ldots, N-1. \tag{15}$$

For each component m, it should be ensured that the chance that the degradation level does not exceed the failure threshold ω_m should be bigger than γ at any inspection, that is

$$\mathrm{Ch}\{Y_{mn} \le \omega_m\} \ge \gamma, \;\; \forall m, n. \tag{16}$$

From a practical point of view, the susceptibility of a component to degradation increases by each maintenance action, so each component cannot be repaired an infinite number of times. Therefore, constraint (17) ensures that for each component m, only a limited number of preventive maintenance actions, $\overline{N}_m$, is allowed, that is

$$\sum_{\theta=n^*+1}^{n} \sigma_{m\theta} \le \overline{N}_m, \quad \forall m, n. \tag{17}$$

where n^* is determined by the following equation:

$$n^* = \begin{cases} \max\{\underset{1\le n'<n}{\operatorname{argmax}}\, \delta_{mn'}\}, & \sum_{\theta=1}^{n} \delta_{m\theta} > 0, \\ 0, & \sum_{\theta=1}^{n} \delta_{m\theta} = 0. \end{cases}$$

Constraint (18) ensures that a preventive maintenance and a replacement maintenance cannot be performed on a component simultaneously

$$\sigma_{mn} + \delta_{mn} \le 1, \quad \forall m, n. \tag{18}$$

Constraint (19) specifies the 0-1 of decision variables as follows:

$$\sigma_{mn}, \delta_{mn} \in \{0, 1\}, \quad \forall m, n. \tag{19}$$

III. Genetic Algorithm

The model established in this paper is a multivariable and nonlinear global optimization problem. An exhaustive examination of all the solution space is not realistic due to the computational time constraint. Advanced meta-heuristic algorithms, such as genetic algorithm, Tabu search, simulated annealing algorithm, and ant colony optimization, are effective approaches to search the global optimal solution in a computationally efficient manner. Among these algorithms, genetic algorithm, which was first proposed by Holland [18] in 1975, has a mature system structure for solving multivariable and nonlinear programming problems, and has been maturely developed and widely used in a wide range of applications, including maintenance optimization problems [19], [20]. Here, a genetic algorithm is designed to solve the proposed multivariable and nonlinear programming model for multicomponent system maintenance plan.

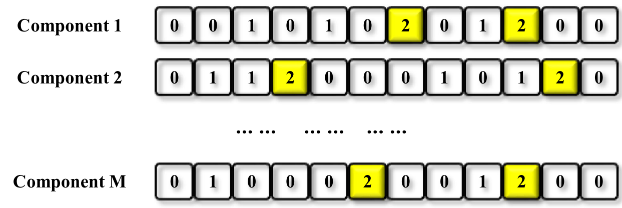


Fig. 1. Structure of chromosomes.

The rules of genetic algorithms are well known, including initialization, fitness function, evaluation, crossover operation, and mutation operation. In this paper, we construct the chromosomes as a two-dimensional (2-D) structure, use roulette wheel to select the chromosomes, self-crossover for the crossover operation, and shift-mutation for the mutation operation.

A. Representation Structure

First of all, we need to establish a one to one correspondence relationship between the solutions and chromosomes. The chromosomes for the proposed model are constructed as a structure of M rows and N columns, as shown in Fig. 1. Each row is a sequence composed of numbers 0, 1, and 2, and corresponds to the maintenance action decision for the component $m, m = 1, 2, \ldots, M$, in which number 0 represents that there is no maintenance action performed at inspection, number 1 represents a preventive maintenance action performed at inspection, and number 2 represents a replacement maintenance action performed at inspection. For example, the series $[0\ 0\ 1\ 0\ 1\ 0\ 2]$ means that a new component is eventually replaced after two preventive maintenance actions.

B. Initialization Operation

An integer pop_size is introduced as the size of population. The initialization operation is to generate a total of pop_size feasible chromosomes as the initialized population. However, for the proposed model, it is difficult to generate a feasible chromosome satisfying the constraint (16). In order to obtain a feasible chromosome more easily, we use the penalty function method to add constraint (16) into the objective function as follows:

$$\min\; C + \pi \sum_{m=1}^{M} \sum_{n=1}^{N} [\mathrm{Ch}\{Y_{mn} \le \omega_m\} - \gamma]^- \tag{20}$$

where π is a very large number, and operator $[x]^+$ means that if $x \ge 0$, take 0; if $x < 0$, take the absolute value of x.

Denote $\mathbf{v} = \{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_M}\}$ as a feasible chromosome, in which $\mathbf{v_m}$ represents the $m\mathrm{th}$ row in it. In order to obtain a feasible chromosome, it needs to respectively generate each feasible row $\mathbf{v_m}$. First, set $i = 1$ and $sum = 0$; Second, randomly select a number r from set $\{0,\ 1,\ 2\}$ for $\mathbf{v_m}(i)$; Third, for the case $r = 0$, set $\mathbf{v_m}(i) = 0$ and $i = i + 1$, and go to the second step. For the case $r = 1$, set $sum = sum + 1$, and then compare the size of sum and $\overline{N}_m$. If $sum \le \overline{N}_m$, set $\mathbf{v_m}(i) = 1$ and $i = i + 1$, and go to the second step; if $sum > \overline{N}_m$, set $\mathbf{v_m}(i) = 2$, $sum = 0$ and $i = i + 1$, and go to the second step. For the case $r = 2$, set $\mathbf{v_m}(i) = 2$, $sum = 0$ and $i = i + 1$, and go to the second step. Repeat the above three steps M times to obtain a feasible chromosome. A total of pop_size feasible chromosomes need to be generated as the initialized population.

C. Fitness Function

Fitness function assigns each chromosome a probability of reproduction, which reflects that its likelihood of being selected is proportional to its fitness relative to the other chromosomes in the population. That is, the chromosomes with higher fitness will have more chance to produce offsprings.

Since the proposed model is a minimization programming one, a chromosome with smaller objective value is better. In this paper, the fitness function is defined as follows:

$$\text{Fitness}(i) = 1 - \frac{f_i}{f_{\max} + f_{\min}}, \; i = 1, 2, \ldots, pop\_size \quad (21)$$

where f_i is the objective value for the ith chromosome, $f_{\max}$ is the maximum objective value for the population, and $f_{\min}$ is the minimum objective value for the population. Obviously, this definition can ensure that the chromosome with smaller objective value can obtain a higher fitness.

D. Selection Operation

The selection operation can improve the quality of the population by assigning a higher probability to chromosomes with a higher fitness to pass to the next generation. In this research, roulette wheel is used for the selection procedure. The roulette wheel is a fitness-proportional selection, where fitter chromosomes are typically more likely to be selected.

The selection operation can be summarized as follows: First, calculate the reproduction probability q_i for each chromosome as follows:

$$q_0 = 0,$$

$$q_i = \sum_{j=1}^{i} \text{fitness of } j\text{th chromosome}, \; i = 1, 2, \ldots, pop\_size;$$

second, generate a random number r in $(0, q_{pop\_size}]$; third, select the chromosome such that $q_{i-1} < r \le q_i$. Repeat the second step and the third step pop_size times to select pop_size chromosomes as the new population.

E. Self-Crossover Operation

Crossover is the main operation to generate diverse chromosomes for the next population. The purpose of the crossover operation is to pass on the superior genes of chromosomes to offsprings and generate a new optimization space. In the traditional genetic algorithm, the crossover operation is usually operated between two chromosomes. However, for the proposed model in this paper, the offsprings produced by the interchromosome crossover operation may be infeasible and violate the maximum number constraint of preventive maintenance actions. Therefore, we use a kind of self-crossover operation in this paper.

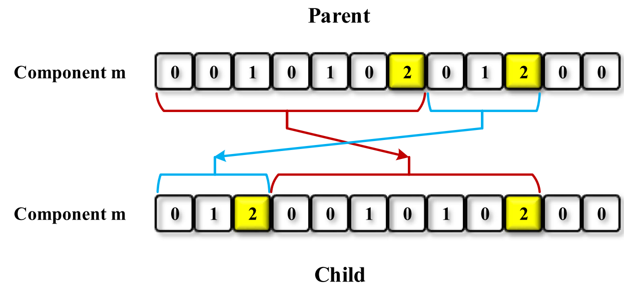


Fig. 2. Self-crossover operation.

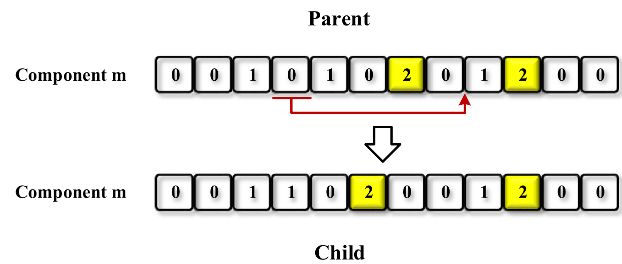


Fig. 3. Shift-mutation operation.

Denote p_c as the crossover probability. The self-crossover operation is expressed as follows. For each chromosome, generate a random number r from $[0, 1]$. If $r < p_c$, select the chromosome as parent. For the parent chromosome, randomly select two life cycles in each row. Exchange the position of the two selected life cycles in each row to generate a child chromosome, as shown in Fig. 2. Obviously, this self-crossover operation can guarantee the feasibility of the generated chromosomes. The offspring generation can be obtained by conducting the self-crossover operation for all the pop_size chromosomes.

F. Shift-Mutation Operation

Mutation is another operation to generate the offsprings, which can increase the probability of escaping from local optima. In this paper, we use a kind of shift-mutation operation. Denote p_m as the probability of mutation. For each chromosome, generate a random number r from $[0, 1]$. If $r < p_m$, select the chromosome as parent. For the parent chromosome, randomly select a gene 0 or 1 in each row, and randomly shift the position of the selected gene in each row to obtain a child chromosome, as shown in Fig. 3. Check the feasibility of child chromosome based on constraint (17). If the child chromosome is feasible, replace the parent chromosome with it; otherwise, keep the parent chromosome. The offsprings generation can be obtained by conducting the shift-mutation operation for all the pop_size chromosomes.

G. General Procedure

After evaluation, selection, crossover, and mutation, we get a new population of chromosomes. After a given number G of iterations of the above steps, the genetic algorithm will terminate.

Algorithm 2 Genetic Algorithm

Step 1. Initialize pop_size chromosomes randomly, and move to **Step 2**.
Step 2. Calculate the fitness values for all chromosomes, and move to **Step 3**.
Step 3. Select the chromosomes by using the roulette wheel, and move to **Step 4**
Step 4. Use self-crossover and shift-mutation operations to update the chromosomes, and move to **Step 5**.
Step 5. Repeat **Step 2** to **Step 5** for G times.
Step 6. Return the chromosome with the best fitness value and decode it to a solution.

TABLE II
MAINTENANCE PARAMETERS

	p_m(\$)	t_m(min)	α_m	β_m	$r_{m\hat{n}}$
1	75000	20	30000	300	[0.7 0.5 0.35 0.2 0.1]
2	130000	40	45000	200	[0.8 0.5 0.45 0.3]
3	50000	15	16500	100	[0.7 0.6 0.5 0.35 0.2 0.1]

The general procedure of the genetic algorithm for this problem is summarized in Algorithm 2.

IV. NUMERICAL EXAMPLES

In this section, we assume a series system with three components, and conduct numerical studies and comparisons based on the presented integrated degradation prediction method and the condition-based maintenance optimization model.

The proposed model contains two contributions: 1) parameters updating method, and 2) maintenance optimization model. In order to analyze and illustrate the two contributions, two cases of numerical examples are conducted in this section. The first case considers a single-component system to describe the effect of parameters updating and to verify the effectiveness of using the RFAD model and parameters updating. The second case considers a 3-component system to illustrate and verify the effectiveness of the proposed model considering opportunistic maintenance.

The parameters in the maintenance optimization model are given in Table II. As mentioned in Table I, the maintenancex effect of each component, $r_{m\hat{n}}$, is degressive. For example, [0.7 0.5 0.35 0.2 0.1] represents the component 1 can be preventively repaired five times at most, and the maintenance effects are 0.7, 0.5, 0.35, 0.2, 0.1, respectively. Other parameters in the model are set as $c_0 = \$300$, $t_0 = 30$ min, $c_d = \$500$. For each component, both the ADT data and monitored data are generating by simulation. Detailed information of the simulation parameters for each component are given in Tables III–V.

TABLE III
INFORMATION OF THE SIMULATION DATA OF COMPONENT 1

Test information	Values
Accelerated stress	Temperature
Stress levels ($°C$)	Normal: 25; ADT: 50, 65, 75
Failure threshold	$\omega = 30$
Inspection interval (hour)	1
Measurement times in ADT	150, 120, 100
Degradation model	$Y(t) = d(S)t + \sigma_B B(t), d(S) \sim N(\mu, \sigma^2)$
Model parameters	$a = 12, b = -5200, \sigma_B = 0.01, \sigma = 0.1\mu$
Sample size in ADT	3

TABLE IV
INFORMATION OF THE SIMULATION DATA OF COMPONENT 2

Test information	Values
Accelerated stress	Temperature
Stress levels ($°C$)	Normal: 25; ADT: 40, 60, 70
Failure threshold	$\omega = 40$
Inspection interval (hour)	1
Measurement times in ADT	240, 170, 120
Degradation model	$Y(t) = d(S)t + \sigma_B B(t), d(S) \sim N(\mu, \sigma^2)$
Model parameters	$a = 11.7, b = -5000, \sigma_B = 0.015, \sigma = 0.1\mu$
Sample size in ADT	4

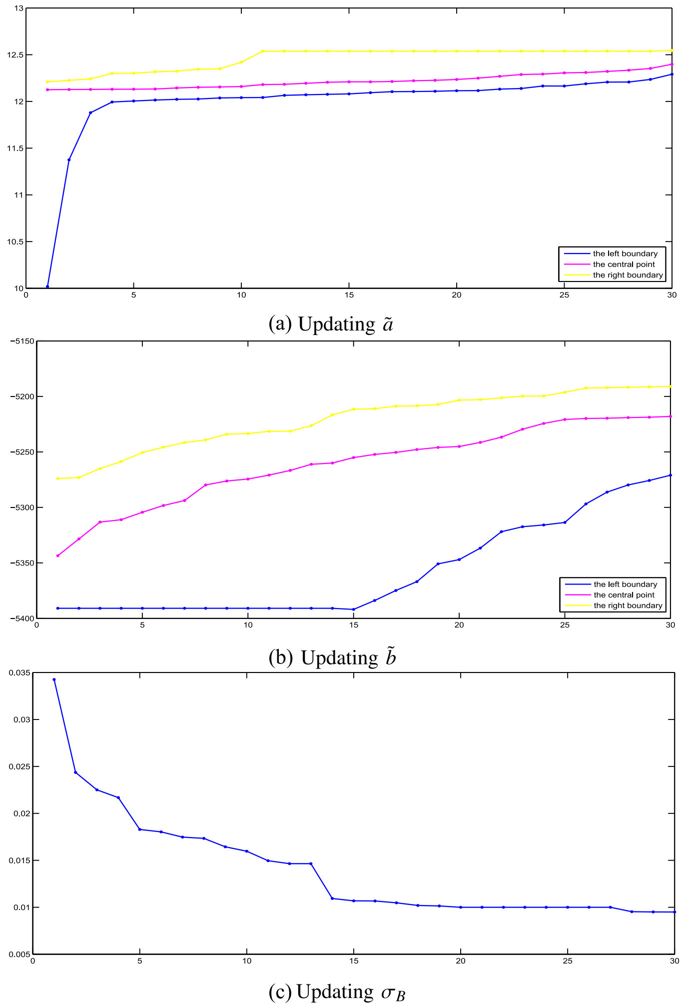
TABLE V
INFORMATION OF THE SIMULATION DATA OF COMPONENT 3

Test information	Values
Accelerated stress	Current
Stress levels (A)	Normal: 1.5; ADT: 4, 7, 9
Failure threshold	$\omega = 30$
Inspection interval (hour)	1
Measurement times in ADT	200, 150, 120
Degradation model	$Y(t) = d(S)t + \sigma_B B(t), d(S) \sim N(\mu, \sigma^2)$
Model parameters	$a = -6, b = 1.4, \sigma_B = 0.015, \sigma = 0.1\mu$
Sample size in ADT	4

A. Case of Single-Component System

In this case, we only consider component 1 as a single-component system to describe the effect of the parameters updating and to verify the effectiveness of using the RFAD model and parameters updating in degradation prediction.

First, we use one time of simulation as an example to describe the effect of the parameters updating. Results are shown in Fig. 4. Fig. 4(a) and (b) illustrates the change of the fuzzy parameters $\tilde{a}$ and $\tilde{b}$, respectively, along with the parameters updating at each inspection. The difference between the left boundary and the right boundary is getting smaller and smaller, and the central point tends to the simulation value. This is because with the increasing number of monitored data collected, the predictions of the fuzzy parameters $\tilde{a}$ and $\tilde{b}$ are more and more precise. Fig. 4(c) shows that the value of parameter σ_B decreases from 0.0345 to 0.0095 with the parameters updating at each inspection. From Fig. 4, we can observe that the changes of the values of parameters are remarkable with the parameters updating at each inspection, so it is essential to update parameters in maintenance optimization.



(a) Updating $\tilde{a}$

(b) Updating $\tilde{b}$

(c) Updating σ_B

Fig. 4. Parameters updating.

Second, we use the proposed maintenance optimization model to verify the effectiveness of using the RFAD model and parameters updating. The *maintenance optimization model based on Wiener process* and the *maintenance optimization model without parameters updating* are also used for comparison.

In the proposed model, the performance degradation of the component are depicted by the RFAD model. In literature, Wiener process is popularly used to model the performance degradation, which considers both the certain time-stress-dependent structure and the random uncertainties caused by the random effects in time dimension. By letting the drift follow a normal probability distribution, the Wiener process model can also take the random uncertainties caused by the unit-to-unit variations into account [21]–[23]. We compare the proposed maintenance optimization model with the *maintenance optimization model based on Wiener process* to illustrate the effectiveness of using RFAD model. Results are shown in Fig. 5. Using the proposed model, there are 6 preventive maintenance actions and 2 replacement maintenance actions performed in the operation horizon, and the total maintenance cost is \$413 523. Using the *maintenance optimization model based on Wiener process*, there are 7 preventive maintenance actions and 3 replacement maintenance actions performed in the operation horizon, and the total maintenance cost is \$525 421. One preventive maintenance action and one replacement maintenance action is added by using the *maintenance optimization model based on Wiener process*, which leads to 27% more expenses than the proposed model. This is because the RFAD model is more conservative than the Wiener process model in the performance prediction [15]. The overestimation of degradation level will increase the number of maintenance actions and lead to waste of residual performance of components.

An integrated degradation prediction framework using both ADT data and monitored data is presented to periodically update the parameters in the proposed model. In order to illustrate the effect of parameters updating, we compare the proposed model with the maintenance optimization model that only uses the ADT data and does not update parameters using the collected monitored data, i.e., *maintenance optimization model without parameters updating*. Results are shown in Fig. 5. It can be observed that using the proposed model, 6 preventive maintenance actions and 2 replacement maintenance actions are performed in the operation horizon, while using the *maintenance optimization model without updating parameters*, 8 preventive maintenance actions and 2 replacement maintenance actions are performed in the operation horizon. Two preventive maintenance actions are added based on the *maintenance optimization model without parameters updating*. This has contributed to the increment of the total maintenance cost. The total maintenance cost based on the proposed model is \$413 523, while the total maintenance cost based on the *maintenance optimization model without parameters updating* is \$465 746, increased by 13%. This is because the proposed model can depict the degradation process of components more accurately, and the capacity of the component can be more fully utilized before its failure.

In summary, the changes of the values of parameters are remarkable with the parameters updating at each inspection, and compared with the *maintenance optimization model based on Wiener process* and the *maintenance optimization model without parameters updating*, the proposed model is more accurate in depicting the degradation process of components. The two other models will erroneously estimate the degradation level, and result in the increment of the total maintenance cost by 27% and 13%, respectively.

B. Case of Multicomponent System

The second case considers the series multicomponent system with three different components to evaluate the performance of the proposed maintenance optimization model. For comparisons, the maintenance optimization model without opportunistic maintenance between more than one components is used.

First, we consider the maintenance optimization model without opportunistic maintenance, that is, the preventive maintenance actions and replacement maintenance actions of each

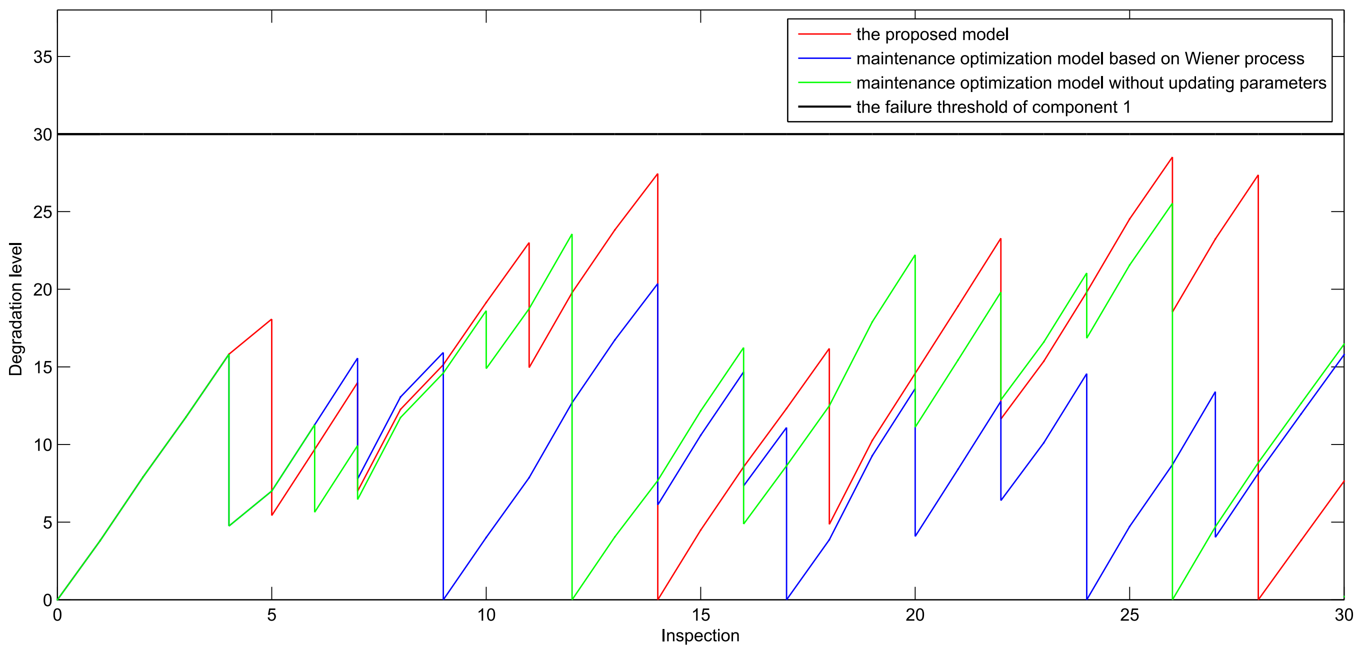


Fig. 5. Comparisons between the proposed model and the maintenance optimization model based on Wiener process/without parameters updating.

component are separately optimized. In this situation, the downtime of the system at each inspection should be the sum of all components, rather than the longest maintenance time among the components. For component 1, the optimized maintenance plan contains 8 preventive maintenance actions and 2 replacement maintenance actions; for component 2, the optimized maintenance plan contains 7 preventive maintenance actions and 2 replacement maintenance actions; for component 3, the optimized maintenance plan contains 9 preventive maintenance actions and 3 replacement maintenance actions. In all, the maintenance plan of the system contains 24 preventive maintenance actions and 7 replacement maintenance actions. The total maintenance cost of the system is \$1 823 829, of which \$1 147 280 is caused by downtime, accounting for 63% of the total maintenance cost, and the direct maintenance cost of the system is \$676 549.

Considering the opportunistic maintenance between more than one components, the downtime of the system at each inspection is the longest maintenance time among the components. In this situation, the total maintenance cost of the system is \$1 439 289, saving maintenance costs for \$384 540, and the cost savings rate is 21%; the downtime loss is \$797 382, accounting for 55% of the total maintenance cost, and reduced \$349 898 compared to the situation without considering the opportunity maintenance; the direct maintenance cost of the system is \$641 907, down by \$34 642. This comparison demonstrates that for multicomponent systems where economic dependence exists, the proposed multicomponent maintenance optimization model is more effective and can lead to lower overall maintenance cost.

Based on the aforementioned results, the downtime loss accounts for a large proportion of the total maintenance cost.

TABLE VI
EFFECT OF THE VALUE OF c_d ON THE MAINTENANCE PLAN

c_d(\$)	No. of preventive maintenance	No. of replacement maintenance	No. of opportunistic maintenance
100	27	5	12
300	24	5	12
500	22	6	14
800	20	7	15
1000	20	7	17

Therefore, the value of the unavailability cost rate of the system, c_d, will greatly affect the maintenance plan of the system. In the following, we investigate the effect of the value of c_d on the maintenance plan of the 3-component system. The results are summarized in Table VI. The results show that the number of preventive maintenance actions is reduced from 27 to 20, while the number of replacement maintenance actions is increased from 5 to 7, as the value of c_d is increased from 100 to 1000. The total number of preventive maintenance actions and replacement maintenance actions is reduced as the value of c_d gets larger. Conversely, the number of opportunistic maintenance actions is increased as the value of c_d gets larger, because the increasing number of opportunistic maintenance actions can reduce the downtime of the system.

V. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

For multicomponent systems, economic dependence exists among different components, and thus it might be cheaper to perform preventive or replacement maintenance actions on multiple components at the same time. In this paper, we focus on the condition-based maintenance optimization for multicomponent systems with consideration of imperfect maintenance. An optimization model has been proposed to minimize the total maintenance cost, including the preventive or replacement maintenance cost and the loss cost incurred by the downtime of the system. An integrated degradation prediction framework using both ADT data and monitored data was presented to timely update the parameters in the proposed model. A genetic algorithm with self-crossover operation and shift-mutation operation was designed to solve the proposed model.

Two cases of numerical examples were conducted to evaluate the performance of the proposed model. The first case considers a single-component system and illustrates that the changes of the values of parameters are remarkable with the parameters updating, and compared with the *maintenance optimization model based on Wiener process* and the *maintenance optimization model without parameters updating*, the proposed model is more accurate in depicting the degradation process of components. The two other models will erroneously estimate the degradation level, and result in the increment of the total maintenance cost by 27% and 13%, respectively. For the case of a 3-component system, the proposed model achieves 21% cost savings rate compared with the one without considering the opportunistic maintenance. Furthermore, the effect of the value of the unavailability cost rate, c_d, on the maintenance plan was investigated. The total number of preventive maintenance actions and replacement maintenance actions is reduced, while the number of opportunistic maintenance actions is increased, as the value of c_d gets larger.

There are some issues that are worthwhile further research. First, this paper has developed a maintenance cost model with linear functional relationships between the reduced degradation level and the maintenance cost/time. Other functional relationships, such as nonlinear, can be developed to accommodate different practical applications. Second, this paper considers only the economic dependence among components. Other kinds of dependence, such as stochastic dependence and structural dependence could also be considered. Third, this paper focus on the nonmonotonic degradation paths. It is also interesting to develop corresponding degradation model and maintenance optimization model with monotonic degradation paths. Lastly, a series multicomponent system is considered in the numerical examples. More applications of the proposed model to other types of systems can be explored, such as parallel systems, combined systems, etc.

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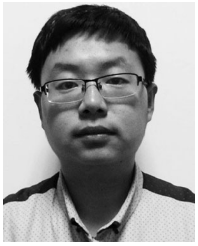

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